

# JAKKI CHAITANYA ESWAR RAJESH

AI Engineer | Azure Data Engineer | Python Developer

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## PROFESSIONAL SUMMARY

Microsoft Certified Azure Data Engineer Associate (DP-203) with 4 years 11 months of hands-on IT experience building production Generative AI systems, voice AI pipelines, and cloud-scale ETL. Lead developer of AppNova AI, an enterprise full-stack multi-agent LLM platform (FastAPI, React/TypeScript, LangGraph, ChromaDB RAG) integrating Azure OpenAI, Claude, Gemini, and Groq. Architected 30+ Azure Data Factory pipelines for U.S. government agencies, cutting data processing time by 50%. Available for remote or Hyderabad-based AI Engineer and Azure Data Engineer roles.

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## TECHNICAL SKILLS

**AI & LLMs:** LangChain, LangGraph, Multi-Agent Orchestration, RAG, Prompt Engineering, Azure OpenAI, OpenAI / GPT, Claude, Gemini, Groq (LLaMA 3), Ollama, Whisper, Hugging Face, Embeddings, NLP, ChromaDB Vector Search

**Cloud & Azure:** Azure Data Factory (ADF), Azure Synapse Analytics, Azure Data Lake Gen2, Azure SQL, Azure Key Vault, Power BI, Azure Logic Apps

**Data Engineering:** ETL / ELT Design, ADF Mapping Data Flows, Cloud Migration (On-Prem to Azure), Data Warehousing, Data Validation, Data Modeling

**Programming:** Python (Advanced), FastAPI, Flask, asyncio, Pydantic, SQL, REST APIs

**AI/ML Libraries:** OpenCV, FFmpeg, MoviePy, scikit-learn, pandas, NumPy, PyTorch, python-docx, GFPGAN, DeblurGANv2

**DevOps & Tools:** Azure DevOps Pipelines, GitHub Actions (CI/CD), Docker, Git, React (TypeScript/Vite), Angular

**Certifications:** Azure Data Engineer Associate — DP-203 (Microsoft)

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## PROFESSIONAL EXPERIENCE

**AI Engineer & Azure Data Engineer | West Advanced Technologies India Pvt. Ltd. | Hyderabad**

Aug 2021 – Present | IT Experience: 4 years 11 months

### AI & GenAI Application Development

- Architected AppNova AI, an enterprise full-stack multi-agent platform (FastAPI + React/TypeScript) with 13 LangGraph agents that analyse legacy codebases and generate modernisation reports as DOCX/PDF; integrated Azure OpenAI, Claude, Gemini, and Groq with automatic fallback across 25+ API keys.
- Refactored the codegen pipeline with parallel chunk waves and deterministic file\_map backfill, increasing build-manifest generation speed by 20%.
- Designed a ChromaDB-backed RAG pipeline with corpus deduplication and AST-based file summaries, retrieving code-relevant context in under 200ms regardless of project size.
- Delivered AriesGPT, a 5-microservice AI platform including a production VoiceBot (Whisper STT → Groq LLaMA 3 → gTTS) with async FastAPI endpoints, structured JSON outputs, and escalation logic.
- Deployed GovGenie, a RAG-powered RFP response generator (LangChain + ChromaDB), reducing bid-writing time by ~70%.
- Automated a resume-to-DOCX system that replaced a 45-minute manual HR task with a 2-minute workflow, and shipped CI/CD via GitHub Actions for zero-downtime deploys.

### Azure Data Factory & ETL Pipeline Engineering

- Orchestrated 30+ production ADF pipelines for U.S. government clients (Contra Costa County, California State Agencies, LA County), ingesting high-volume RMS records into Azure SQL Database and Data Lake with zero data loss.
  - Accelerated data processing by 50% versus the prior manual batch process using Mapping Data Flows, Lookup activities, and parameterized Linked Services.
  - Reduced manual data-review effort by 40% by automating validation, deduplication, and transformation logic, with real-time pipeline health via Azure Monitor and Power BI reporting.
  - Led the ARIES cloud migration of a 40+ table criminal-justice system (multi-million rows) from on-prem SQL Server to Azure with a full lineage audit trail.
  - Maintained Linked Services, Integration Runtimes, and self-hosted IR configurations across Azure SQL, SFTP, REST, and Blob sources with secure credentials via Azure Key Vault.
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## CAREER TRANSITION & PRIOR WORK HISTORY

*The following section documents my background before entering the IT industry in 2021, included for employment-history completeness. These are non-IT positions.*

### Career Transition & Self-Directed Learning | Dec 2018 – Jul 2021 (2 years 8 months)

- Pursued self-directed study in Python, SQL, and cloud data engineering, earning the Microsoft Azure Data Engineer Associate (DP-203) certification — the foundation for joining WATI in August 2021.

## Office Co-Ordinator | Eastern Petro Technologies India Pvt. Ltd. | Kakinada | Apr 2018 – Nov 2018

Non-IT role — EPFO verified

- Managed quotations, purchase orders, and cross-departmental coordination reporting to the CEO.

## Process Associate | Macray Technologies Pvt. Ltd. | Kakinada | Oct 2016 – Mar 2017

Non-IT role — earliest career position

- Handled U.S. solar-market cold calling and customer acquisition for Home Solar Solutions, West Palm Beach, Florida.
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## KEY PROJECTS

### AppNova AI — Enterprise Multi-Agent Code Modernisation Platform | FastAPI · React/TypeScript · LangGraph · Azure OpenAI · Claude · Gemini · ChromaDB · SSE | 2025 – Present

- Pioneered a 13-agent LangGraph DAG that leverages Claude Code to convert legacy codebases into runnable stacks with audits, reports, and zero Claude Max API cost.
- Scaled throughput with SHA-256 content-fingerprint caching and SSE streaming, letting repeated uploads skip all LLM calls and serve cached results in seconds.

### AriesGPT — Angular + FastAPI AI Monorepo (Law-Enforcement Media Workflows) | Angular · FastAPI · GFPGAN · DeblurGANv2 · Whisper · ViT-GPT2 · Groq | 2025

- AI-powered law enforcement and investigation platform unifies voice, crime intelligence, crime mapping, facial/vehicle deblur, and video narrative, built on Angular + FastAPI.
- Streamlined an end-to-end voice assistant (Whisper STT → Groq LLaMA 3 → gTTS) with async FastAPI, confidence thresholds, and escalation handling for failed transcription.

### GovGenie — AI-Powered RFP Response Generator | FastAPI · Ollama (LLaMA 3.1) · LangChain · ChromaDB · Hugging Face | 2025

- An AI assistant automating government RFP response generation and resume-to-template population, running locally with an LLM for cost-effective, quick results.

### Video to Narrative — Surveillance Video Summarisation App | Flask · OpenCV · Whisper · ViT-GPT2 · Groq (LLaMA 3) | 2025

- Converted surveillance footage into law-enforcement incident summaries via OpenCV frame extraction, ViT-GPT2 captioning, Whisper transcription, and Groq LLaMA 3 JSON output.

### AI Content Bot — Fully Automated Daily AI/Tech Video Pipeline | Python · Gemini 2.5 Flash · Edge TTS · MoviePy · FFmpeg · YouTube & Instagram APIs · GitHub Actions | 2025 – Present

- Automated a daily pipeline that turns trending RSS news into a 1080×1920 captioned video and publishes hands-free to YouTube Shorts and Instagram Reels via GitHub Actions cron.
- Added a Gemini auto-fix agent that, on pipeline failure, analyses the job log and source files and opens a patched pull request with no manual intervention.

### ModernizeAI — Multi-Stack AI App Generator | Angular 18 · FastAPI · Ollama · Streamlit · scikit-learn | 2025

- Ollama-powered full-stack app with Angular, optional Streamlit UI, and FastAPI backend generates and packages starter apps, data, and validation.
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## EDUCATION & CERTIFICATIONS

- B.Tech, Mechanical Engineering | GIET College of Engineering, Rajahmundry | May 2015
  - Microsoft Certified: Azure Data Engineer Associate (DP-203) | Microsoft
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## GOVERNMENT CLIENT ENGAGEMENTS (VIA WEST ADVANCED TECHNOLOGIES)

- **Contra Costa County, California** — ADF pipelines for RMS data migration; 30+ entity tables synced between legacy on-prem SQL Server and Azure SQL Database.
- **California State Agencies** — Parameterized ADF pipelines for real-time reporting and compliance data across multiple agencies from a single template.
- **Los Angeles County** — BEAGLE system optimization and ARIES cloud migration; improved query performance, compliance audit trails, and documented DR/rollback.
- **WATI (Government Contractor)** — GovGenie RFP automation; reduced proposal preparation from 5+ days to under 24 hours for standard government bids.